

TITLE

Science-based Management of Introduced Mountain Goats on Kodiak, Alaska

SUMMARY

The introduced mountain goat (*Oreamnus americanus*) population in Kodiak National Wildlife Refuge is growing exponentially and has reached a critical threshold. If densities exceed carrying capacity, the population will degrade resources and crash. As a non-native species, a high density population of goats has the potential to adversely impact important Refuge resources. The shared management goal of the Kodiak Refuge and the Alaska Department of Fish and Game (ADF&G) is to avoid irreversible resource damage while maintaining goat abundances at levels that satisfy hunter harvest expectations. To achieve this goal, managers need to understand the relationships between the goat population, harvest levels and their habitat. Quantifying these relationships requires detailed empirical data on mountain goat abundances, population dynamics, resource selection patterns, and diets. Through applied research, we will: (1) develop a monitoring method to provide refined and cost-effective estimates of mountain goat abundances and distributions; (2) provide managers with critical information on the relationship between population limiting factors (including hunting) and mountain goat survival, natality and recruitment; and (3) deliver a thorough analysis of mountain goat resource selection patterns. Together, the results from this study will be used to support a model that successfully relies on hunter harvest to maintain the population at a level that avoids negative impacts to Refuge resources.

BACKGROUND AND JUSTIFICATION

In 1952, eighteen mountain goats were introduced to Kodiak Island, Alaska. By 2014, the population had grown exponentially to over 3,000 goats. If left unchecked, this population will likely degrade habitat, overshoot carrying capacity and crash to low levels, before increasing to a theoretical carrying capacity below peak abundance (Caughley 1970). During the initial growth phase, the high-density population could cause detrimental landscape-level effects by altering structure and composition of native plant communities and functional integrity of soils (Hobbs 1996, Spear and Chown 2009). The impacts of a high-density population of introduced mountain goats on the vegetation, and other wildlife, are potentially significant, as documented elsewhere (Houston and Schreiner 1994, Aho and Weaver 2003, Lemke 2004). Alpine plant communities are highly prone to impacts associated with overuse because of their limited distribution, restricted growing season, and shallow soils (Courchamp et al. 2003). On Kodiak Island, impacts from goats on such plants could be especially severe because it is an isolated island with little potential for recolonization, and because its temperate climate supports a short growing season that is less resilient to disturbance (Courchamp et al. 2003). Ecologically cascading impacts from a high density mountain goat population on Kodiak are likely because goats compete with native species, including Kodiak brown bear, for alpine forage resources (Atwell et al. 1980, Cobb et al. 2012).

Managing mountain goats and their habitats on Kodiak Island is a high priority of the Refuge and ADF&G, as reflected in agency policies and plans. Managing non-native species to minimize impacts on native resources while continuing to provide harvest

opportunities is a goal in the Refuge Comprehensive Conservation Plan (CCP) (FWS 2006). To meet this goal, the CCP identified two specific objectives: monitoring trends in mountain goat distributions, sizes and productivity; and quantifying mountain goat habitat preferences to improve understanding of their influence on habitat conditions. These goals, objectives, and concerns are further supported by the Refuge's Inventory and Monitoring Plan, in which both mountain goat population monitoring and research on goat resource selection identified as high priority, "selected" projects (FWS 2014). Similar to the Refuge, ADF&G has identified four explicit information needs related to management of mountain goats on Kodiak Island: low cost methods to index annual population trends; improved understanding of movements and habitat selection; a harvest plan to provide ample hunter opportunities while maintaining habitat conditions; and improved understanding of impacts to habitats (Van Daele and Crye 2008).

To prevent irreversible impacts to native habitat resources and harvest opportunities, managers need objective information on goat population size, the effects of hunter harvest and other factors on annual changes in abundance, and goat resource selection patterns. Results from this study are crucially needed to be able to quantify density-dependent impacts from mountain goats, and to develop an associated empirical harvest management model maintains mountain goat abundances at a level that avoids negative impacts to native flora and fauna, thereby meeting management objectives.

OBJECTIVES

- 1) Develop a mountain goat aerial survey sightability model that accurately estimates detection probabilities, population sizes, and biases associated with aerial surveys.*
- 2) Estimate mountain goat reproduction and survival rates to develop a data-driven harvest management model.*
- 3) Quantify mountain goat seasonal and annual habitat selection.*

Objective 1: Sightability Model and Population Estimation— Obtaining reliable estimates of mountain goat population abundances and trends are challenging because of the species' cryptic behavior and the rugged habitats they utilize. Monitoring mountain goat population abundances on Kodiak Island has been especially problematic because of the large size of the survey area (>2,000 km²) and frequent occurrence of unsuitable survey weather. Current survey efforts provide a measure of minimum population size, which is inadequate for modeling the effects of recreational sport harvest on population abundance. To date, there has been no effort to adjust population estimates relative to sightability biases or to quantify statistical confidence surrounding population estimates. Previous studies have determined that aerial observers generally detect only 60-70% of mountain goats present (Cichowski et al. 1994, Poole et al. 2000, Pauley and Crenshaw 2006, Poole 2007). Improved methods and statistical techniques that explicitly incorporate detection probability into estimates of animal abundance have mediated many problems associated with surveying mountain ungulates (Udevitz et al. 2006, Rice et al. 2009, Jenkins et al. 2011). These analytical techniques present a promising, but untested, means to estimate mountain goat abundance on Kodiak that could be used with a

sampling approach to provide managers with a cost-effective means to generate critical information needed for effective harvest management.

Objective 2: Population Dynamics— To forecast changes in abundance, managers require knowledge of population vital rates and factors (including hunting) that drive these rates. The population dynamics of mountain goats are not well understood, especially for introduced populations, and vital rates can vary among regions (Hamel et al. 2006, Festa-Bianchet and Cote 2008).

Mountain goat populations typically are regulated by physical and environmental factors that drive other northern ungulate populations (Smith 1986, White et al. 2011), but the extent and relative degrees to which these factors influence population dynamics are unclear. Mountain goats have lower recruitment and delayed primiparity compared to other ungulates (Smith 1986, Festa-Bianchet and Cote 2008), and are therefore sensitive population limiting factors (Hamel et al. 2006). Potential limiting factors for mountain goats on Kodiak include hunting, winter severity (Adams and Bailey 1982, White et al. 2009, White et al. 2011), forage availability (Pfitsch and Bliss 1985, Houston and Stevens 1988), availability of escape terrain, disease (Kerr and Holmes 1966, Dunbar et al. 1986, Froelich 2000, Gross 2001), and predation. Currently, the relationship between Kodiak mountain goat population dynamics and limiting factors, especially hunting, is unclear because we lack data on their survival, recruitment, and causes of death.

Objective 3: Habitat Selection— To evaluate the relationship between mountain goat abundance and their impacts to forage resources, we need information on their habitat selection patterns. Data on mountain goat habitat selection are rare compared to other ungulate species, partly because mountain goats inhabit steep, rugged mountainous environments characterized by extreme climate conditions. Directly monitoring mountain goats during nutritionally stressful periods such as winter, and periodic events such as parturition, is especially challenging. However, understanding habitat requirements during these critical time periods is crucial to linking landscape needs to population dynamics. Habitat selection data is also needed to better understand the relative impacts goat may be having on specific habitats and plant communities during certain seasons, and thereby allow managers to focus future conservation efforts. Mountain goats in some regions of Kodiak Island exhibit seasonal shifts in their use of vegetation types and elevations (Hjeljord 1970, Smith and Van Daele 1987, Cobb et al. 2013), but results are primarily limited to spring and summer seasons. Habitat selection patterns during winter are unclear, but understanding these are important because mountain goats typically exhibit more concentrated space use patterns during this period, which could translate to more substantial range impacts. Combined with a recently completed diet and feeding site selection studies (Cobb et al. 2013), a crucial opportunity exists to integrate field-based habitat data with high-resolution mountain goat spatial-use data collected via GPS-linked radio-collars. Integrating these data sets would enable a detailed assessment of the spatial distributions and habitat condition of mountain goat summer and winter range and, thereby, provide managers with empirical information needed to assess seasonal range use.

METHODS

Capture and Radio-collar— Adult mountain goats ($n = 30/\text{year}$, 2015-2017) will be captured using standard helicopter darting techniques (Taylor 2000, White et al. 2012) and immobilized by injecting carfentanil citrate via projectile syringe fired from a dart gun (Cap-Chur, Douglasville, GA). During handling, we will monitor all animals following standard veterinary procedures (Taylor 2000, White et al. 2012). We will collect biological samples (blood, tissue, pellets, and hair) and morphological data (mass, and horn and body dimensions). Following handling, we will reverse the immobilizing agent with naltrexone hydrochloride. All capture procedures will be approved by the FWS Animal Care and Use Committee.

We will fit captured mountain goats with a GlobalStar GPS radio telemetry collar (Advanced Telemetry Systems, Inc., Isanti, MN) and an ear tag. We will program collars to collect location data at a 12-hour interval (collar lifetime: 6-8 years), which will be emailed daily from the collars to the Refuge biologist. We will post-process and filter the GPS collar (D'Eon and Delparte 2005).

Objective 1: Population Estimation— We will conduct annual population abundance and composition surveys from a fixed-wing aircraft during August. Surveys will be flown along established flight paths between 1,500-4,500 ft above sea level and follow geographic contours. During surveys, both the pilot and experienced observer will classify all mountain goats observed by age class (adults or kids). After groups are sighted and classified, we will determine whether radio-collared animals are in observed groups using image-stabilizing binoculars.

We will estimate the abundance of mountain goats using Bayesian procedures that involve statistically integrating survey-specific mark-resight estimates and modeled covariate-based survey-level estimates (White and Pendleton 2011). During aerial surveys, we will collect data to evaluate individual- and survey-level “sightability”. For accomplishing survey-level objectives, we will enumerate the number of radio-collared animals seen during surveys and compare this value to the total number of radio-collared animals present in the area surveyed. To gather individual-based “sightability” data, we will characterize behavioral, environmental and climatic conditions for each radio-collared animal seen and not seen (i.e. missed) during surveys. We will locate missed radio-collared animals use radio-telemetry to gather associated covariate information for these animals. Since observers will have general knowledge of where specific individual radio-collared animals will likely reside (i.e. ridge systems, canyon complexes, etc.), we expect to locate missed animals within 5-15 minutes after an area is initially surveyed.

Objective 2: Population Dynamics— We will estimate kidding success and first-year kid survival by monitoring radio-collared animals monthly via a fixed-wing aircraft equipped for radio-telemetry tracking. During aerial surveys, we will observe radio-collared adult female mountain goats (typically using 14X image-stabilizing binoculars) to determine whether they gave birth, and if so, how long kids survived. We will analyze these data to derive estimates of mountain goat kid survival following established methods (Testa 2004, Hebblewhite and Merrill 2011).

We will determine mortality rates of GPS-collared mountain goats by monitoring changes in collar accelerometer data, which collars will email daily to the Refuge wildlife biologist. When mortalities are detected, we will attempt to investigate sites as soon as possible to ascertain causes of death and collect relevant biological samples. We will estimate annual survival of radio-collared animals using known-fate models in Program MARK (White 1999), which allow for staggered entry of animals after the start of the study. We will quantify the relative importance of individual vital rates to annual changes in population abundance by calculating sensitivity and elasticity values. We will incorporate the results from known-fate models into stochastic population projection models to determine how mountain goat abundances vary in response to different levels of population limiting factors, such as hunting. These models will enable managers to examine the effects of different harvest strategies on mountain goat population dynamics, and make informed manage decisions.

Objective 3: Habitat Selection—To identify selectively-used habitats, we will develop resource selection function (RSF) models (i.e., Boyce 2002) by overlaying GPS collar location data and remotely-sensed covariate spatial data in a GIS framework. We will analyze these data using logistic regression to derive selection coefficients for each covariate, by individual GPS-collared animal. We will compute the average inter-individual coefficient value (and confidence interval) for each covariate using a two-stage modeling framework (Fieberg et al. 2010), stratified by summer and winter seasons. We will validate the predictive performance of RSF models using k-fold cross-validation (Boyce et al. 2002). We will use the results of RSF models to identify selectively-used habitats annually and by season. Ultimately, such habitat models will be combined with habitat-specific plant biomass and nutritional quality data (Cobb et al. 2013) to estimate mountain goat nutritional carrying capacity at specific sites across Kodiak.

PRODUCTS

1. An improved and cost-effective method of monitoring abundance of the mountain goat population, which includes sightability corrections and confidence intervals.
2. A predictive population model that describes the relative contributions of hunting and other potential limiting factors on goat survival and reproductive rates.
3. A predictive GIS map of annual and seasonal resource selection, based on empirical data from radio-collared goats.

MANAGEMENT IMPLICATIONS

SCHEDULE

FY 2015

- Monitor existing radio collared goats for survival and reproductive success
- Capture and collar 30 adult mountain goats in July. Begin monitoring collared goats for survival, parturition, and resource selection patterns.
- Refine statistical methodology and field protocols for sightability estimation.
- Establish mountain goat survey units and survey mountain goats in August.
- Submit progress report.

FY 2016

- Capture and collar 30 adult mountain goats in July.
- Continue monitoring collared goats for survival, parturition and resource selection patterns.
- Survey mountain goats in survey units in August. Refine statistical methodology and field protocols for sightability estimation.
- Obtain updated remotely-sensed data of mountain goat ranges.
- Begin developing resource selection and population dynamics models.
- Submit progress report.

FY 2017

- Refine resource selection and population projection models.
- Submit final report.

PERSONNEL

Principle Investigator: McCrea Cobb, Ph.D., Wildlife Biologist, U.S. Fish and Wildlife Service, Kodiak National Wildlife Refuge, Kodiak, Alaska 99615
Phone: 907-487-0246, fax: 907-487-2630, email: mccrea_cobb@fws.gov

Co-investigator: Bill Pyle, Supervisory Wildlife Biologist, U.S. Fish and Wildlife Service, Kodiak National Wildlife Refuge, Kodiak, Alaska 99615
Phone: 907-487-0228, fax: 907-487-2630, email: bill_pyle@fws.gov

Co-investigator: Nathan Svoboda, Wildlife Biologist, Alaska Department of Fish and Game, Kodiak, Alaska 99615
Phone: 907-486-1880, email: nathan.svoboda@alaska.gov

BIBLIOGRAPHY

- Adams, L. G., and J. A. Bailey. 1982. Population dynamics of mountain goats in the Sawatch Range, Colorado. *The Journal of Wildlife Management* 46:1003-1009.
- Aho, K., and T. Weaver. 2003. Classification of alpine plant communities of the Northern Rocky Mountain Volcanics. Report to Yellowstone National Park. YELL-NPS-5119. 82 pp.
- Atwell, G., D. L. Boone, J. Gustafson, and D. B. Vernon. 1980. Brown bear summer use of alpine habitat on the Kodiak National Wildlife Refuge. *Bears: Their Biology and Management* 4:297-305.
- Boyce, M. S., P. R. Vernier, S. E. Nielsen, and F. K. A. Schmiegelow. 2002. Evaluating resource selection functions. *Ecological Modelling* 157:281-300.
- Caughley, G. 1970. Eruption of ungulate populations, with emphasis on Himalayan thar in New Zealand. *Ecology* 51:53-72.
- Cichowski, D. B., D. Haas, and G. Schultze. 1994. A method used for estimating mountain goat numbers in the Babine Mountains Recreation Area, British Columbia. *Biennial Symposium of the Northern Wild Sheep and Goat Council* 9:56-64.
- Cobb, M., B. Pyle, and H. Helling. 2013. Summer diet, feeding site selection, and forage resources of mountain goats on Kodiak Island, Alaska. Refuge Report no. 2013.7. Kodiak National Wildlife Refuge, U.S. Fish and Wildlife Service, Kodiak, AK.

- Cobb, M. A., H. Helling, and B. Pyle. 2012. Summer diet and feeding site location patterns of an irrupting mountain goat population on Kodiak Island, Alaska. Biennial Symposium of the Northern Wild Sheep and Goat Council 18:122-135.
- Courchamp, F., J. L. Chapuis, and M. Pascal. 2003. Mammal invaders on islands: impact, control and control impact. *Biological Reviews* 78:347-383.
- D'Eon, R. G., and D. Delparte. 2005. Effects of radio-collar position and orientation on GPS radio-collar performance, and the implications of PDOP in data screening. *Journal of Applied Ecology* 42:383-388.
- Dunbar, M. R., W. J. Foreyt, and J. F. Evermann. 1986. Serologic evidence of respiratory syncytial virus infection in free-ranging mountain goats (*Oreamnus americanus*). *Journal of Wildlife Diseases* 22:415-416.
- Festa-Bianchet, M., and S. D. Cote. 2008. Mountain goats: ecology, behavior and conservation of an alpine ungulate. Island Press, Washington, DC.
- Fieberg, J., J. Matthiopoulos, M. Hebblewhite, M. S. Boyce, and J. L. Frair. 2010. Correlation and studies of habitat selection: problem, red herring or opportunity? *Philosophical Transactions of the Royal Society B-Biological Sciences* 365:2233-2244.
- Froelich, K. 2000. Viral diseases of northern ungulates. *Rangifer* 20:83-97.
- FWS. 2006. Revised comprehensive conservation plan and environmental impact statement: Kodiak National Wildlife Refuge. U.S. Fish and Wildlife Service, Kodiak, AK.
- _____. 2014. Inventory and monitoring plan for Kodiak National Wildlife Refuge. Kodiak, Alaska.
- Gross, J. E. 2001. Evaluating effects of an expanding mountain goat population on native bighorn sheep: a simulation model of competition and disease. *Biological Conservation* 101:171-185.
- Hamel, S., S. D. Cote, K. G. Smith, and M. Festa-Bianchet. 2006. Population dynamics and harvest potential of mountain goat herds in Alberta. *Journal of Wildlife Management* 70:1044-1053.
- Hebblewhite, M., and E. Merrill. 2011. Demographic balancing of migrant and resident elk in a partially migratory population through forage-predation tradeoffs. *Oikos* 120:1860-1870.
- Hjeljord, O. 1970. Feeding ecology and habitat preferences of mountain goat in the Kodiak and Kenai mountains. University of Alaska, Fairbanks.
- Hobbs, N. T. 1996. Modification of ecosystems by ungulates. *Journal of Wildlife Management* 60:695-713.
- Houston, D. B., and E. G. Schreiner. 1994. Mountain goats in the Olympic National Park: biology and management of an introduced species. *National Park Service Science Monograph* 25.
- Houston, D. B., and V. Stevens. 1988. Resource limitation in mountain goats - a test by experimental cropping. *Canadian Journal of Zoology-Revue Canadienne De Zoologie* 66:228-238.
- Jenkins, K. J., K. Beirne, R. Hoffman, and W. Baccus. 2011. Mountain goat abundance and population trends in the Olympic Mountains, Washington, 2011. USGS Open File Report 2011-1313. Reston, VA.

- Kerr, G. R., and J. C. Holmes. 1966. Parasites of mountain goats in west central Alberta. *Journal of Wildlife Management* 30:786-790.
- Lemke, T. O. 2004. Origin, expansion, and status of mountain goats in Yellowstone National Park. *Wildlife Society Bulletin* 32:532-541.
- Pauley, G. R., and J. G. Crenshaw. 2006. Evaluation of paintball, mark-resight surveys for estimating mountain goat abundance. *Wildlife Society Bulletin* 34:1350-1355.
- Pfitsch, W. A., and L. C. Bliss. 1985. Seasonal forage availability and potential vegetation limitations to the mountain goat population, Olympic National Park. *American Midland Naturalist* 113:109-121.
- Poole, K. G. 2007. Does survey effort influence sightability of mountain goats (*Oreamnos americanus*) during aerial surveys? *Wildlife Biology* 13:113-119.
- Poole, K. G., D. C. Heard, and G. S. Watts. 2000. Mountain goat inventory in the Robson Valley, British Columbia. *Biennial Symposium of the Northern Wild Sheep and Goat Council* 12:114-124.
- Rice, C. G., K. J. Jenkins, and W.-Y. Chang. 2009. A sightability model for mountain goats. *Journal of Wildlife Management* 73:468-478.
- Smith, C. A. 1986. Rates and causes of mortality in mountain goats in southeastern Alaska. *Journal of Wildlife Management* 50:743-746.
- Smith, R. B., and L. J. Van Daele. 1987. Terror lake hydroelectric project, final report on mountain goat studies. Alaska Department of Fish and Game. Kodiak, AK. 38 pp.
- Spear, D., and S. L. Chown. 2009. Non-indigenous ungulates as a threat to biodiversity. *Journal of Zoology* 279:1-17.
- Taylor, W. P. 2000. Wildlife capture and restraint manual. Alaska Department of Fish and Game, Anchorage, AK.
- Testa, J. W. 2004. Population dynamics and life history trade-offs of moose (*Alces alces*) in south-central Alaska. *Ecology* 85:1439-1452.
- Udevitz, M. S., B. S. Shults, L. G. Adams, and C. Kleckner. 2006. Evaluation of aerial survey methods for Dall's sheep. *Wildlife Society Bulletin* 34:732-740.
- Van Daele, L. J., and J. R. Crye. 2008. Unit 8 mountain goat management report. Pages 112-129 in P. Harper, editor. Mountain goat management report of survey-inventory activities 1 July 2005 - 30 June 2007. Alaska Department of Fish and Game., Juneau, AK.
- White, G. C. 1999. Program MARK: survival estimation from populations of marked animals. *Bird Study* 46:120-138.
- White, K. S., D. P. Gregovich, G. W. Pendleton, N. L. Barten, R. Scott, A. Crupi, and D. N. Larsen. 2012. Mountain goat population ecology and habitat use along the Juneau Access road corridor, Alaska. Final wildlife research report, ADF&G/DWC/WRR-2012-02. Alaska Department of Fish and Game, Juneau, AK.
- White, K. S., G. W. Pendleton, D. Crowley, H. J. Giese, K. J. Hundertmark, T. McDonough, L. Nichols, M. Robus, C. A. Smith, and J. W. Schoen. 2011. Mountain goat survival in coastal Alaska: effects of age, sex, and climate. *The Journal of Wildlife Management* 14:1-12.

White, K. S., G. W. Pendleton, and E. Hood. 2009. Effects of snow on Sitka black-tailed deer browse availability and nutritional carrying capacity in southeastern Alaska. *Journal of Wildlife Management* 73:481-487.



Figure 1. Original mountain goat introduction site (1953), and current estimated range of expansion (2010), Kodiak Island, Alaska.

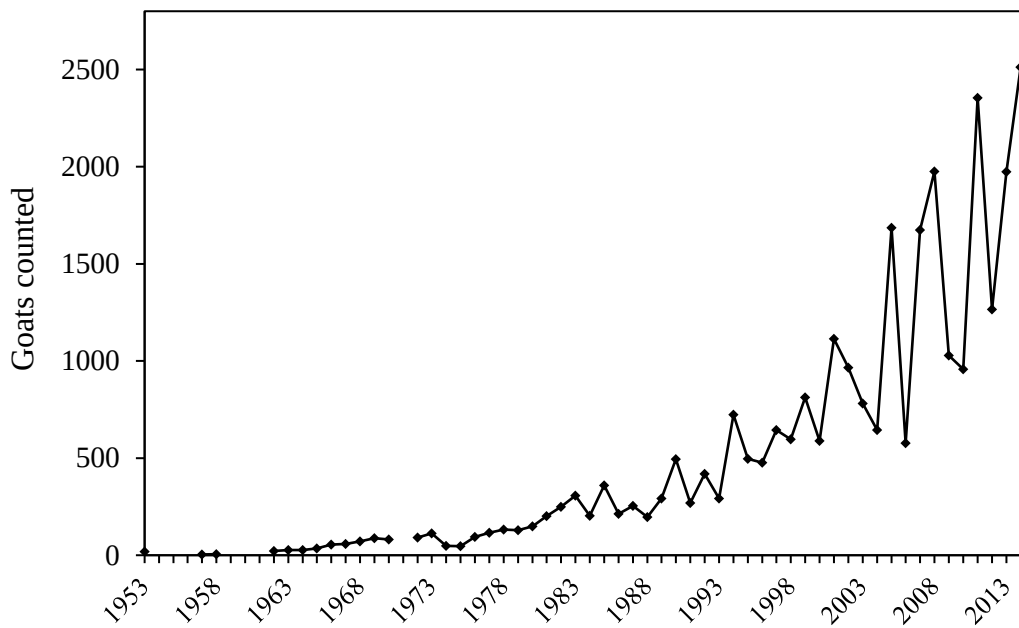


Figure 2. Number of mountain goats counted during aerial surveys conducted by the Alaska Department of Fish and Game and U.S. Fish and Wildlife Service, Kodiak Island, Alaska, 1953 - 2014. Data from annual counts vary because of variable survey effort. Counted from these surveys should be considered a minimum estimate of mountain goat abundance.

Project budget.

		<u>FY 2015</u>				<u>FY 2016*</u>		
Category	Description	\$/unit	Refuge	ADF&G	NRPC request	Refuge	ADF&G	NRPC Request
<i>Personnel:</i>								
Refuge wildlife biologist (GS-11)	Lead captures, analyze data	\$36	\$14,400			\$30,240		
ADF&G wildlife biologists	Assist with captures and monitoring	\$32		\$9,600			\$10,080	
ADF&G wildlife biologist	Travel and lodging	\$1,500		\$1,500			\$1,575	
<i>Capture:</i>								
GPS collars	ATS GlobalStar GPS collar	\$700	\$21,000			\$22,050		
GlobalStar subscription	Annual cost/collar	\$195	\$5,850			\$6,143		
Veterinary supplies	Drugs, reversal, etc.	\$115	\$3,450			\$3,623		
Helicopter	Daily on-call rate	\$2,450			\$17,150			\$18,008
Helicopter	Hourly, 0.85 hrs/goat	\$375			\$9,563			\$10,041
Jet fuel	35 gal/hr @ \$6.25/gal	\$219			\$5,578			\$5,578
Transport fuel and personnel	Beaver, 4 hrs @ \$280/hr	\$280	\$1,120			\$1,176		
Av gas	Beaver, 88 gal @ \$6.25/gal	\$550	\$550			\$578		
Refuge pilot (GS-12)	4 hrs @ \$40/hr	\$160	\$160			\$168		
Misc	Food, supplies for capture crew	\$2,000	\$2,000			\$2,100		
<i>Radio Telemetry Monitoring:</i>								
Airplane (Top Cub)	(4 hrs @ \$125/hr)*12/year	\$6,000			\$6,000			\$6,300
Av gas	(40 gal @ \$6.25/gal)*12/year	\$3,000			\$3,000			\$3,150
Refuge pilot (GS-12)	(4 hrs @ \$40/hr) * 12/year	\$1,920	\$1,920			\$2,016		
TOTAL:			\$50,450	\$11,100	\$41,291	\$68,093	\$11,655	\$43,076

*FY 2016 adjusted for 5% annual inflation